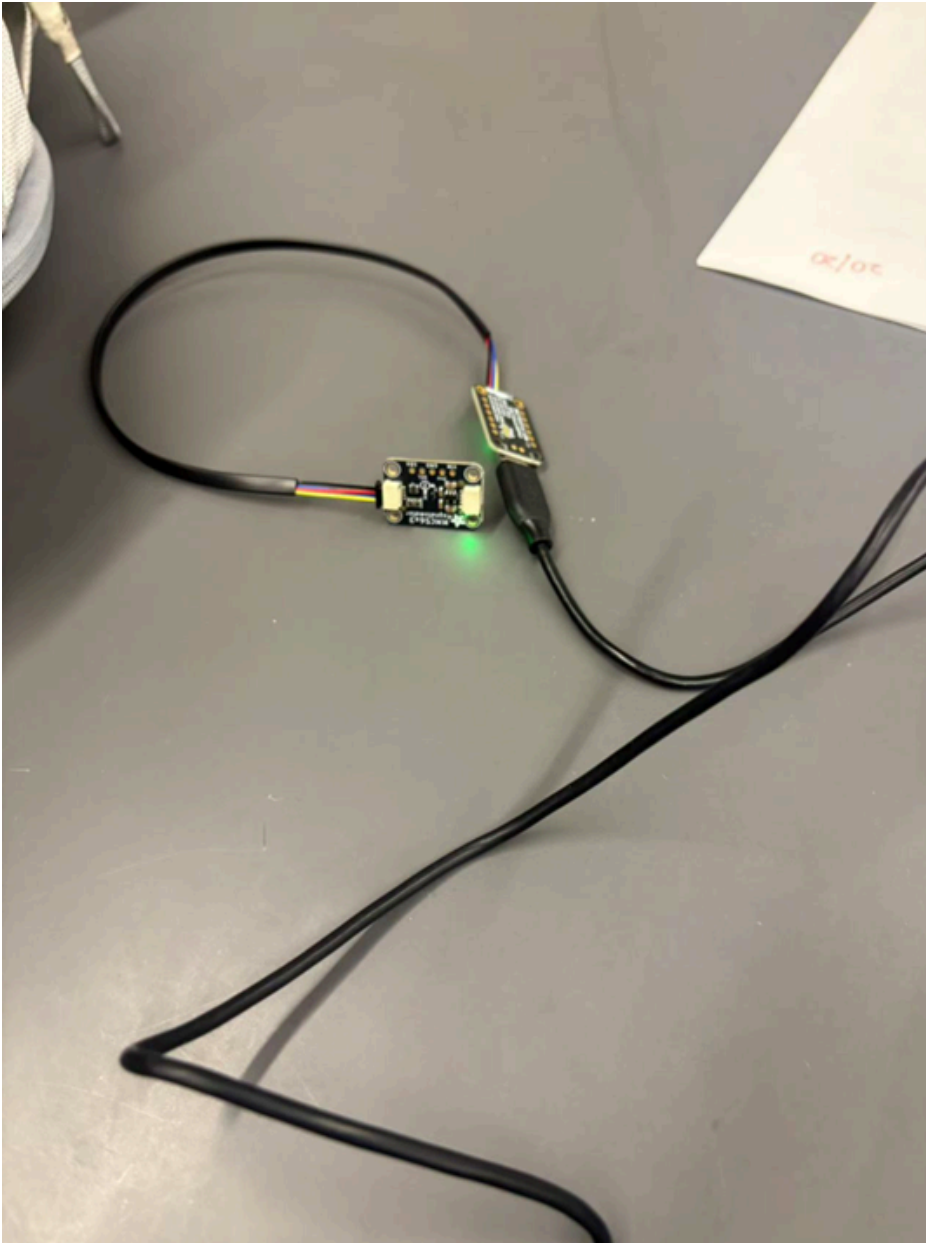


B_field



MMC5603 Magnetometer

MCP2221A Breakout - General Purpose USB
to GPIO ADC I2C

calculate board offset with rotation

B *I* U $\div$  Aa    ... 

```
Degaussing...

--- CALIBRATION MODE ---
Slowly rotate the sensor in all 3D directions (Figure-8 pattern).
Press CTRL+C when the numbers stop changing.

Current Bias -> X: 12.50 | Y: 19.71 | Z: -86.72 uT

--- CALIBRATION COMPLETE ---
Copy these numbers into your main tracking code:
OFFSET_X = 12.50
OFFSET_Y = 19.71
OFFSET_Z = -86.72
```

calculate board offset with 90 degree rotations

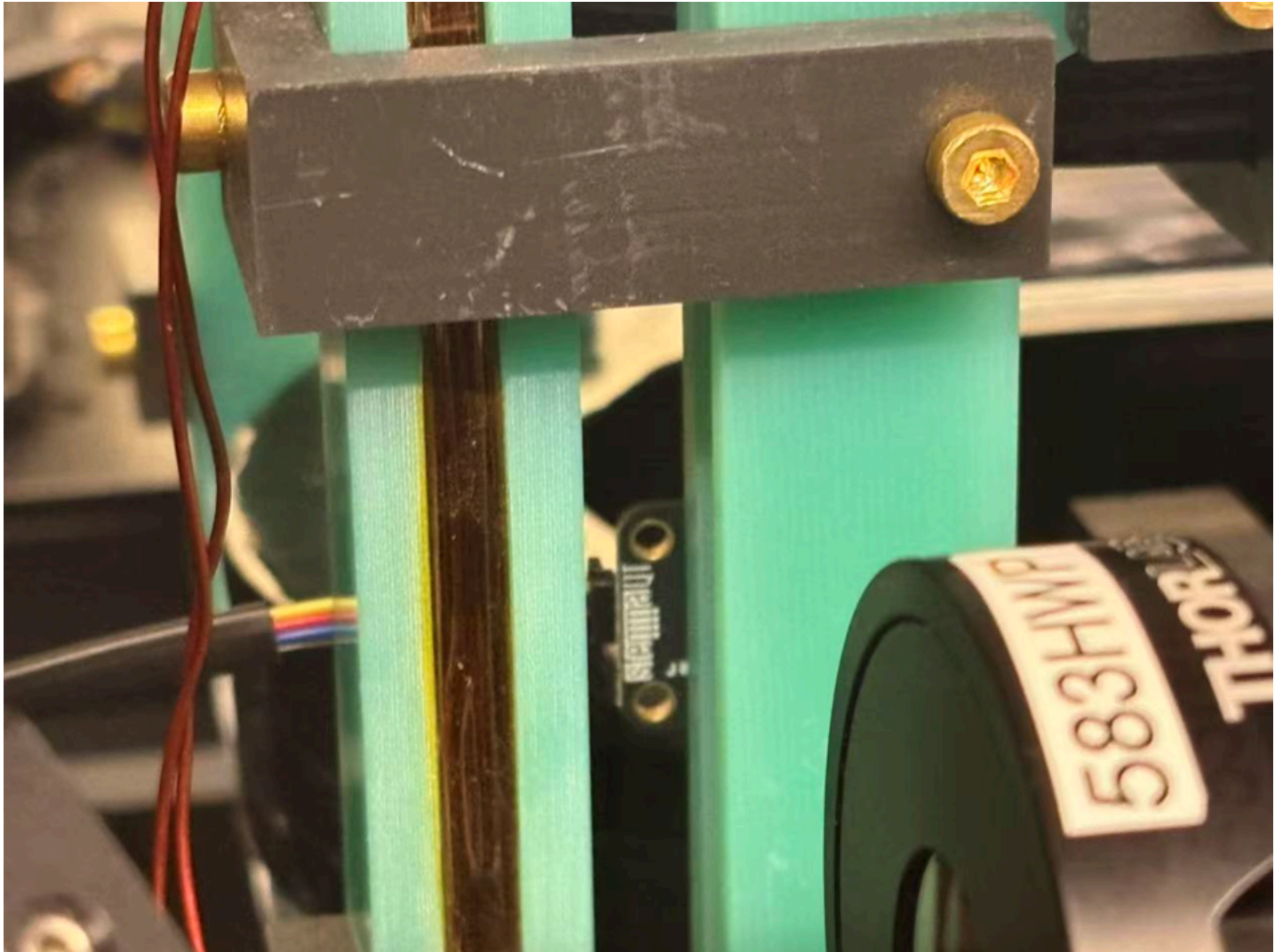
FINAL GROUND TRUTH BIAS RESULTS

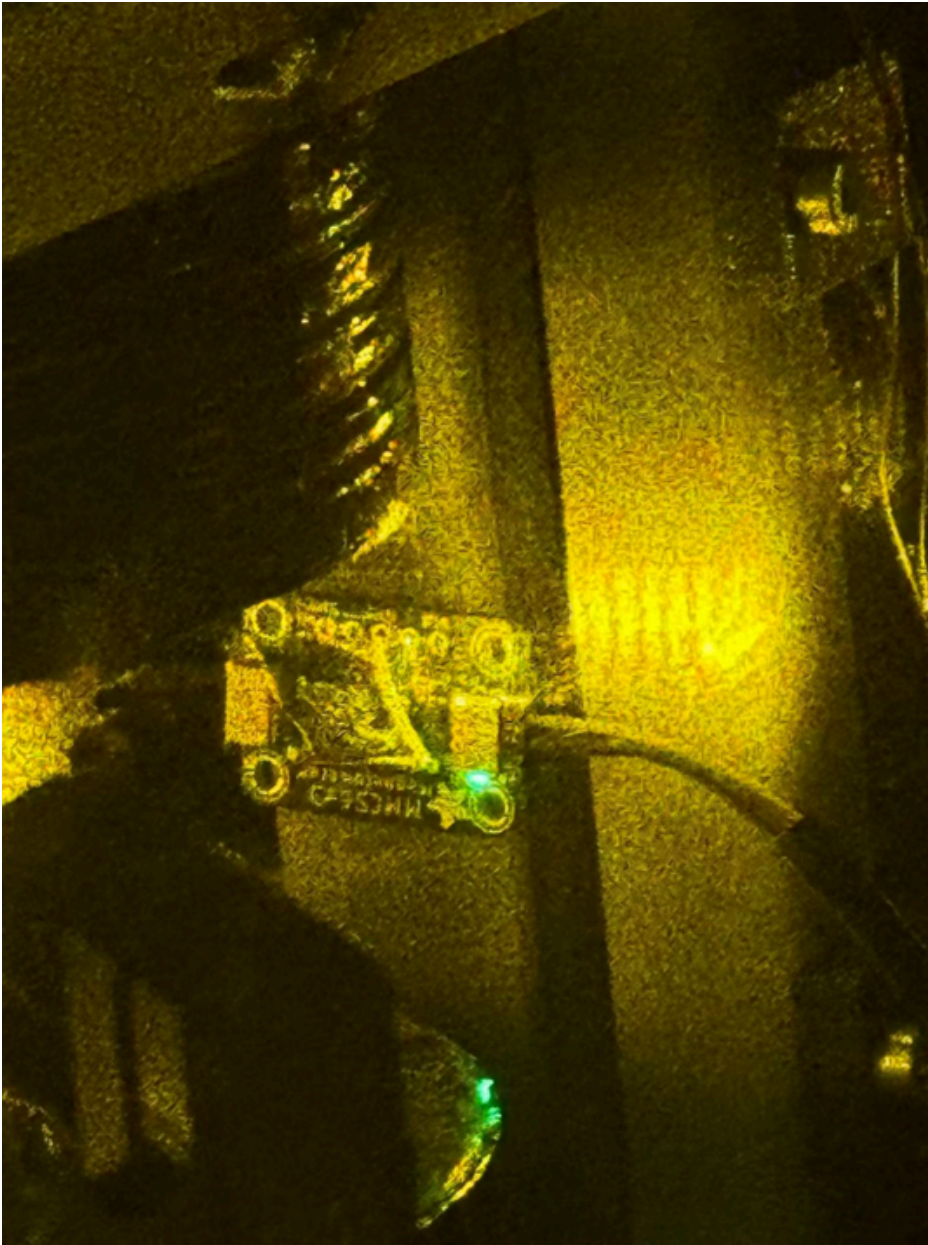
Use these values in your compensation code:

OFFSET_X = 12.360

OFFSET_Y = 18.990

OFFSET_Z = -78.235





preliminary installation

Z=facing wall from door

X=down

Y=left

- $\text{sensor_x} = -\text{REALz} \rightarrow \text{REALz} = -\text{sensor_x}$
- $\text{sensor_y} = \text{REALx} \rightarrow \text{REALx} = \text{sensor_y}$
- $\text{sensor_z} = -\text{REALy} \rightarrow \text{REALy} = -\text{sensor_z}$

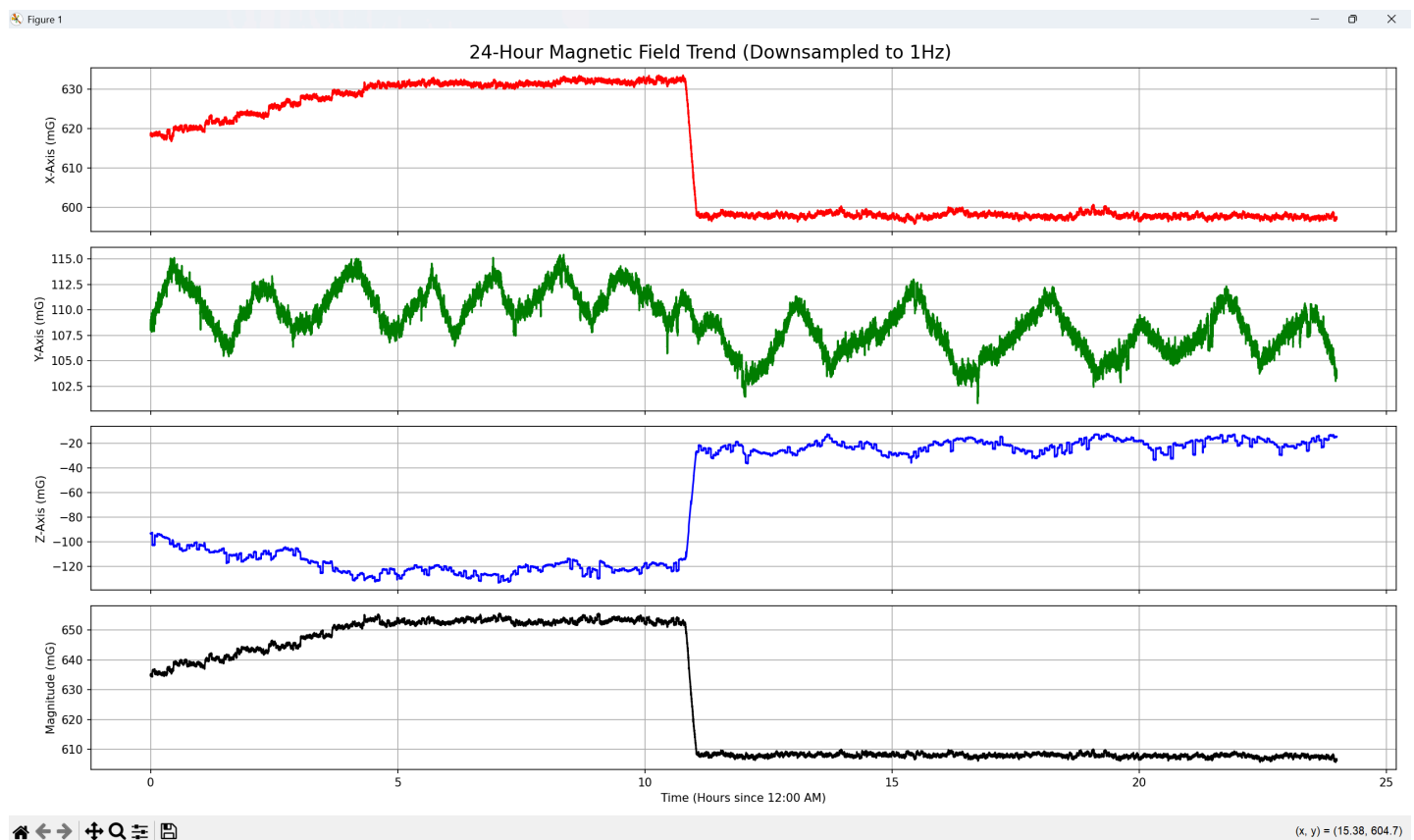
LAB coordinates:

- Y is out toward to lab door
- Z is up
- X is left looking from door

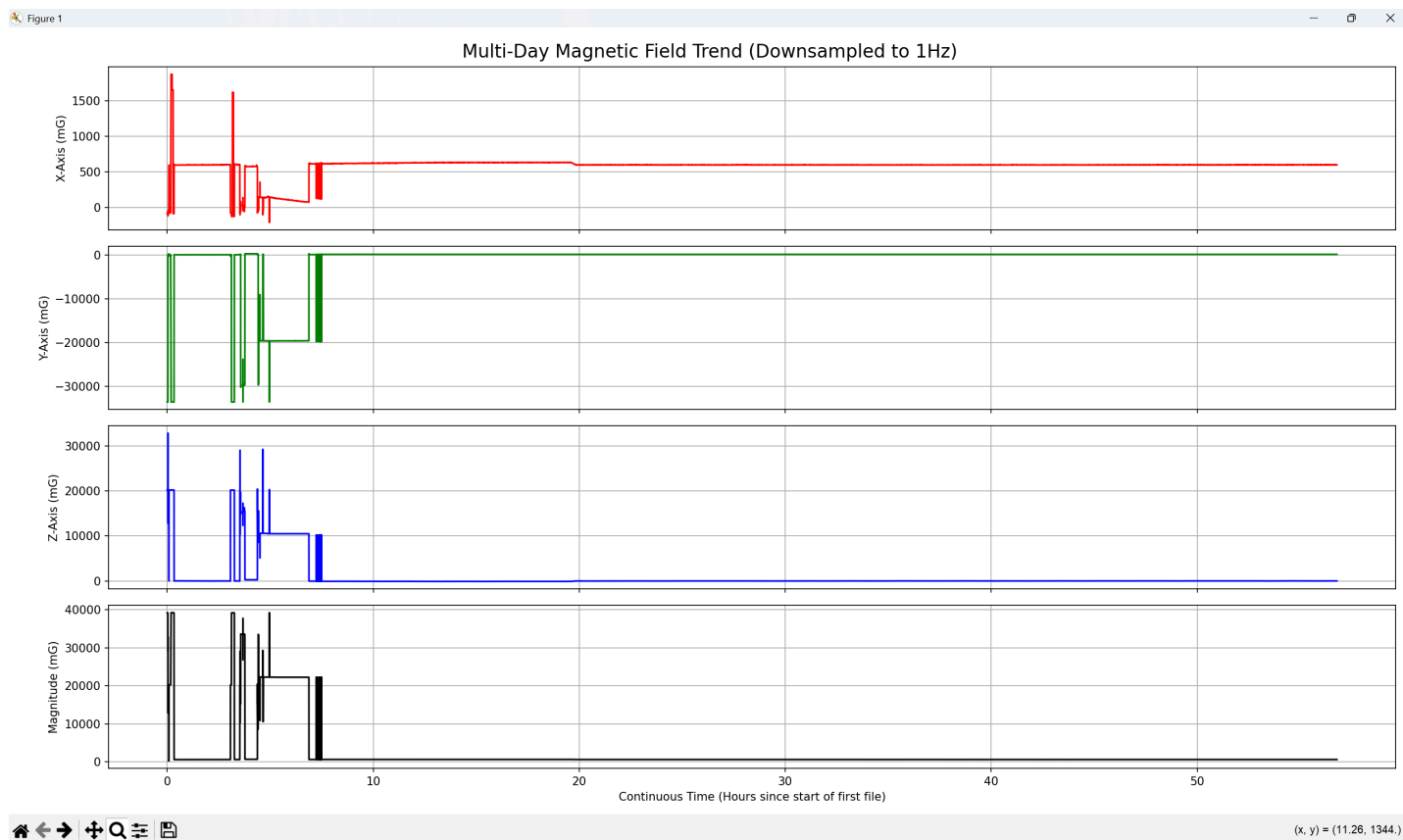
Current latency:

- 4ms
sensor hardware
- 1ms
broadcast to UDP localhost
- 1ms PSU

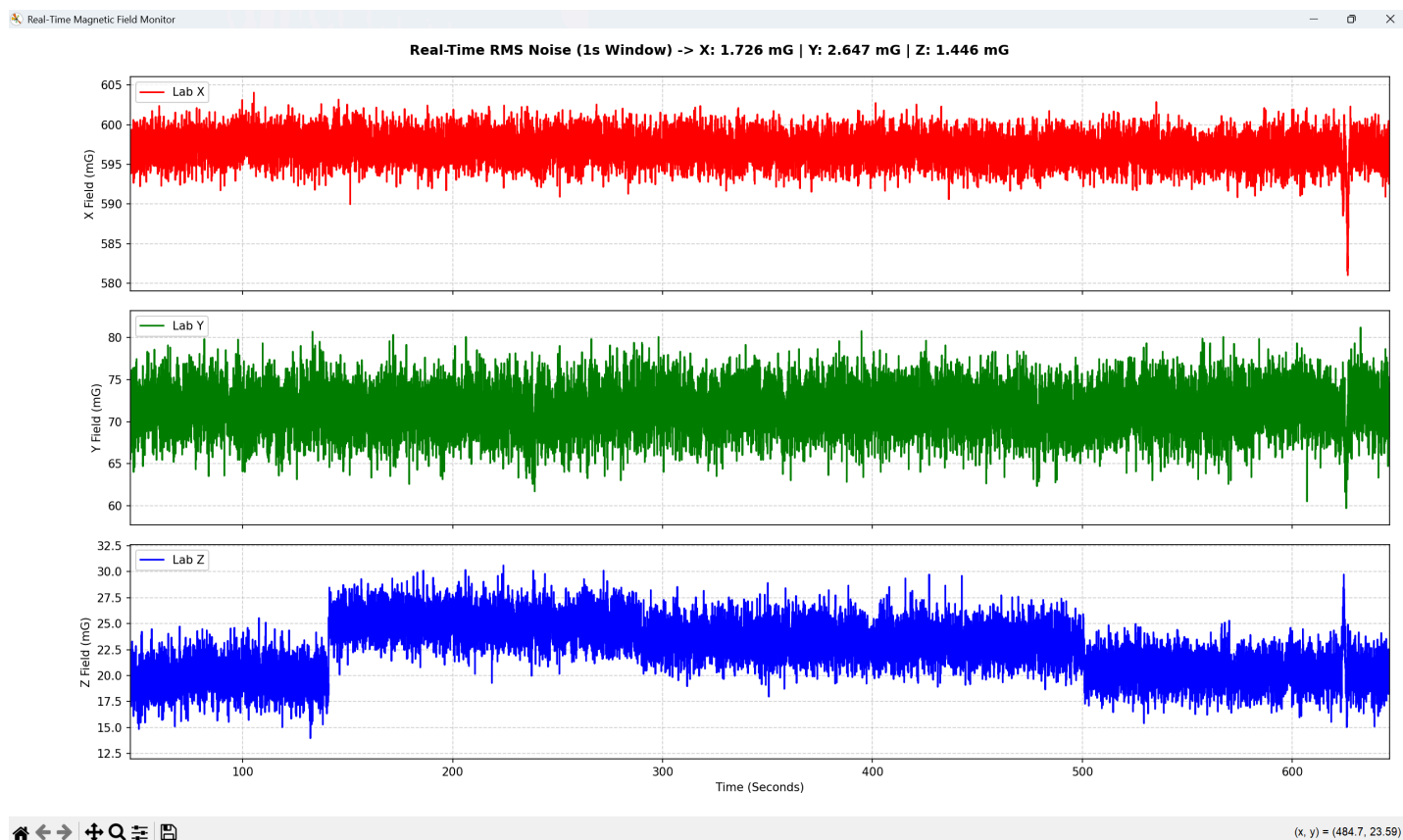
2026/4/14



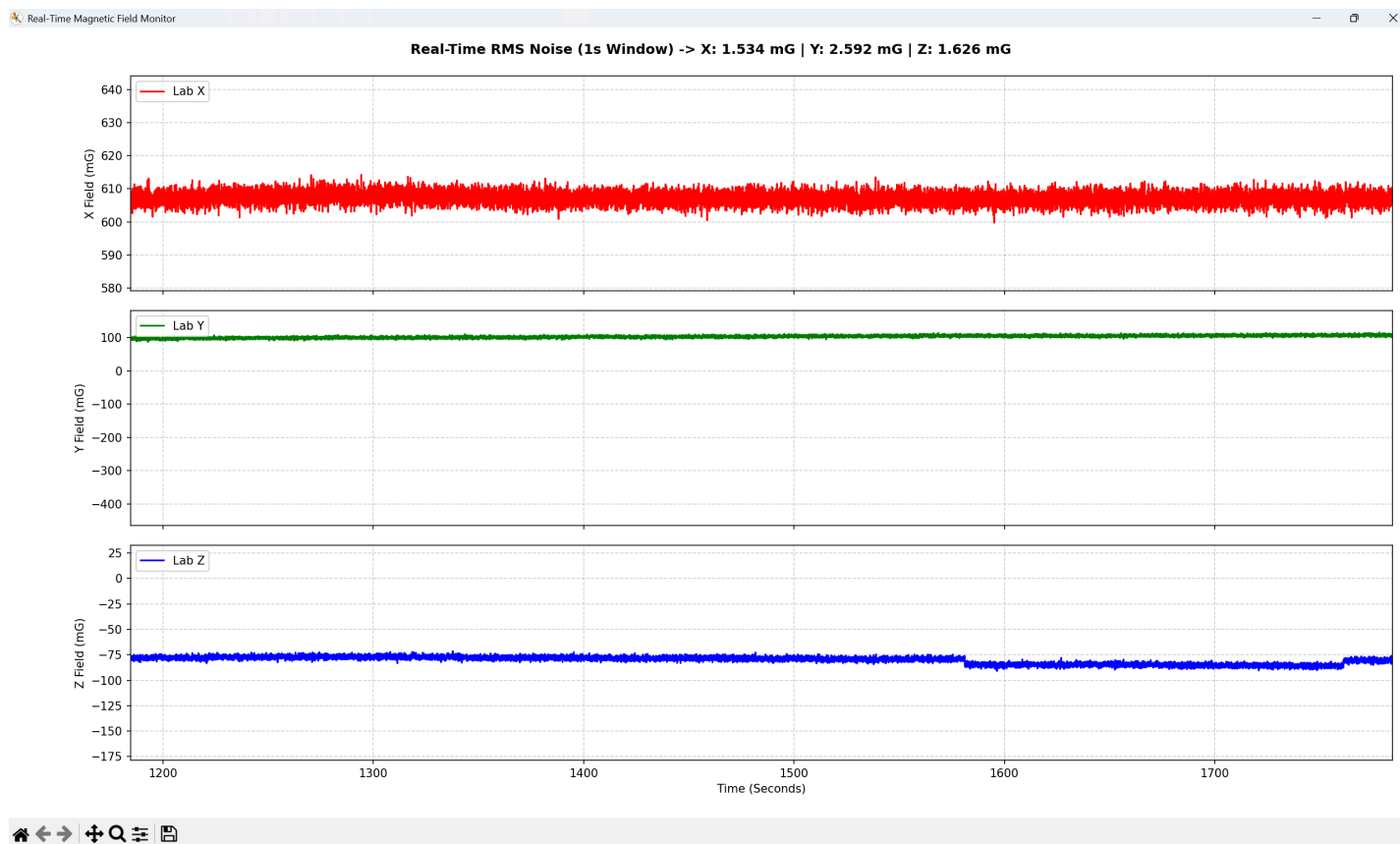
Saturday



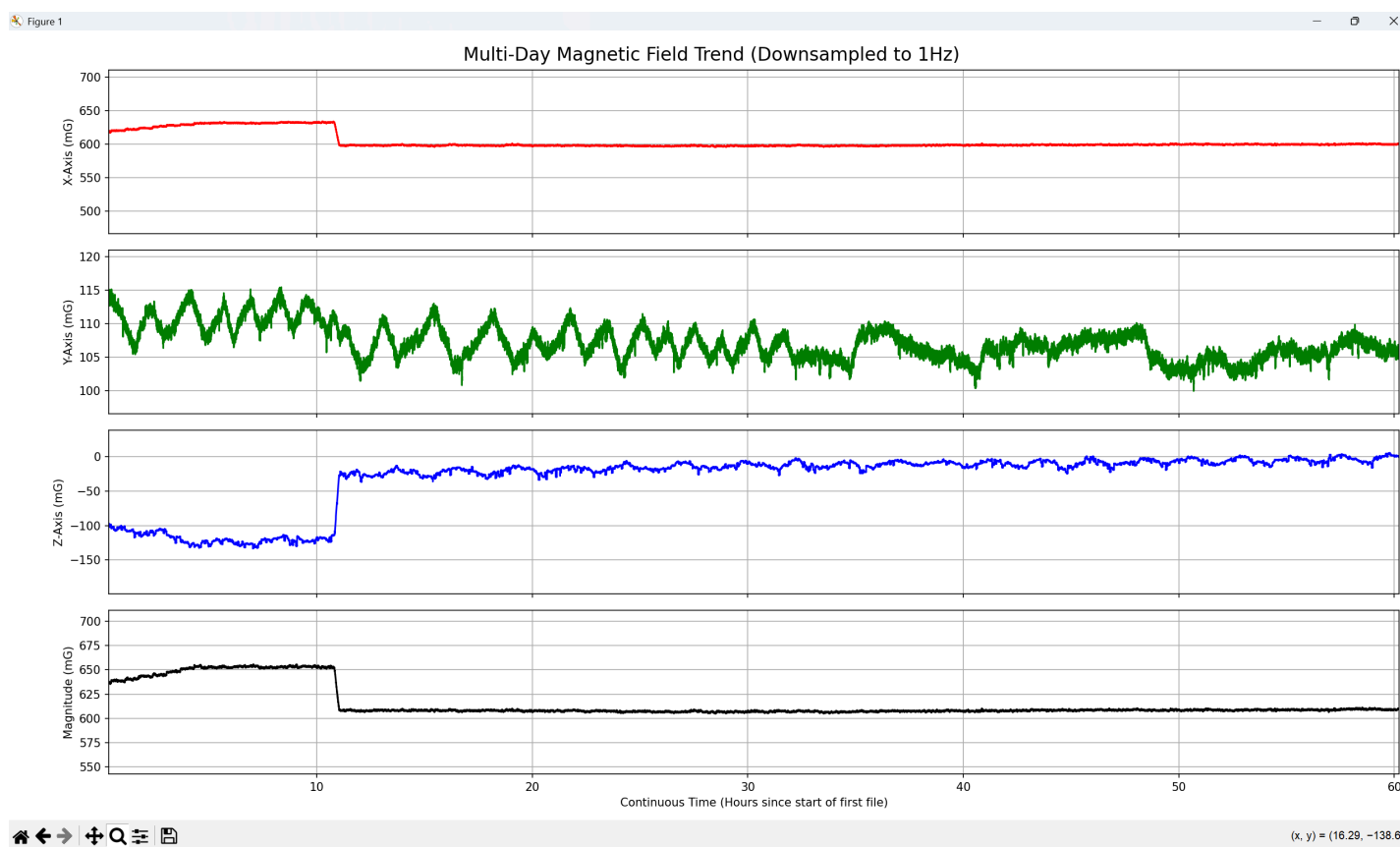
Friday 2pm, Saturday, Sunday Large field coming from Bias and MOT coils



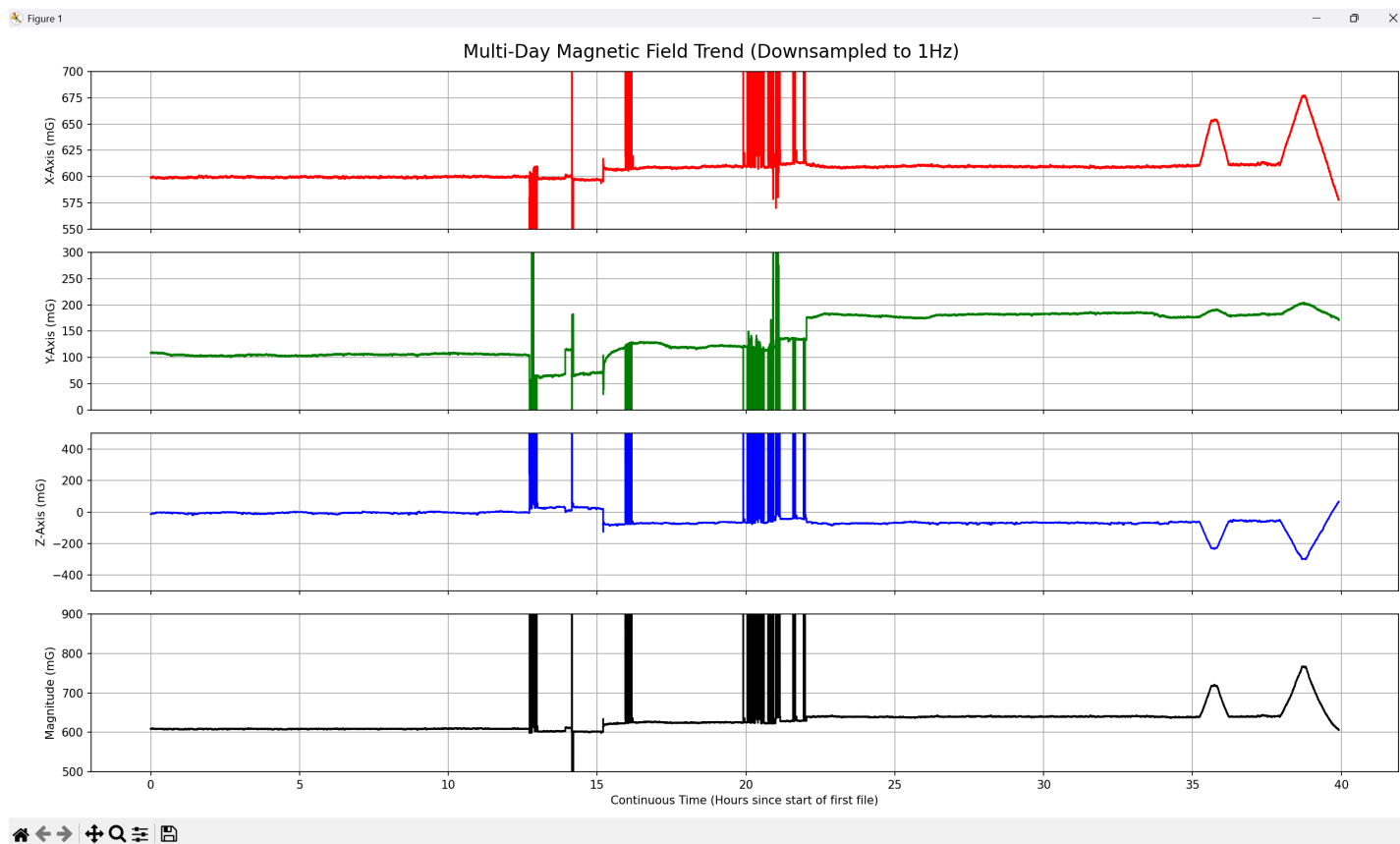
Real time display of data script, this is when a airpod case was put near the glass cell and taken out



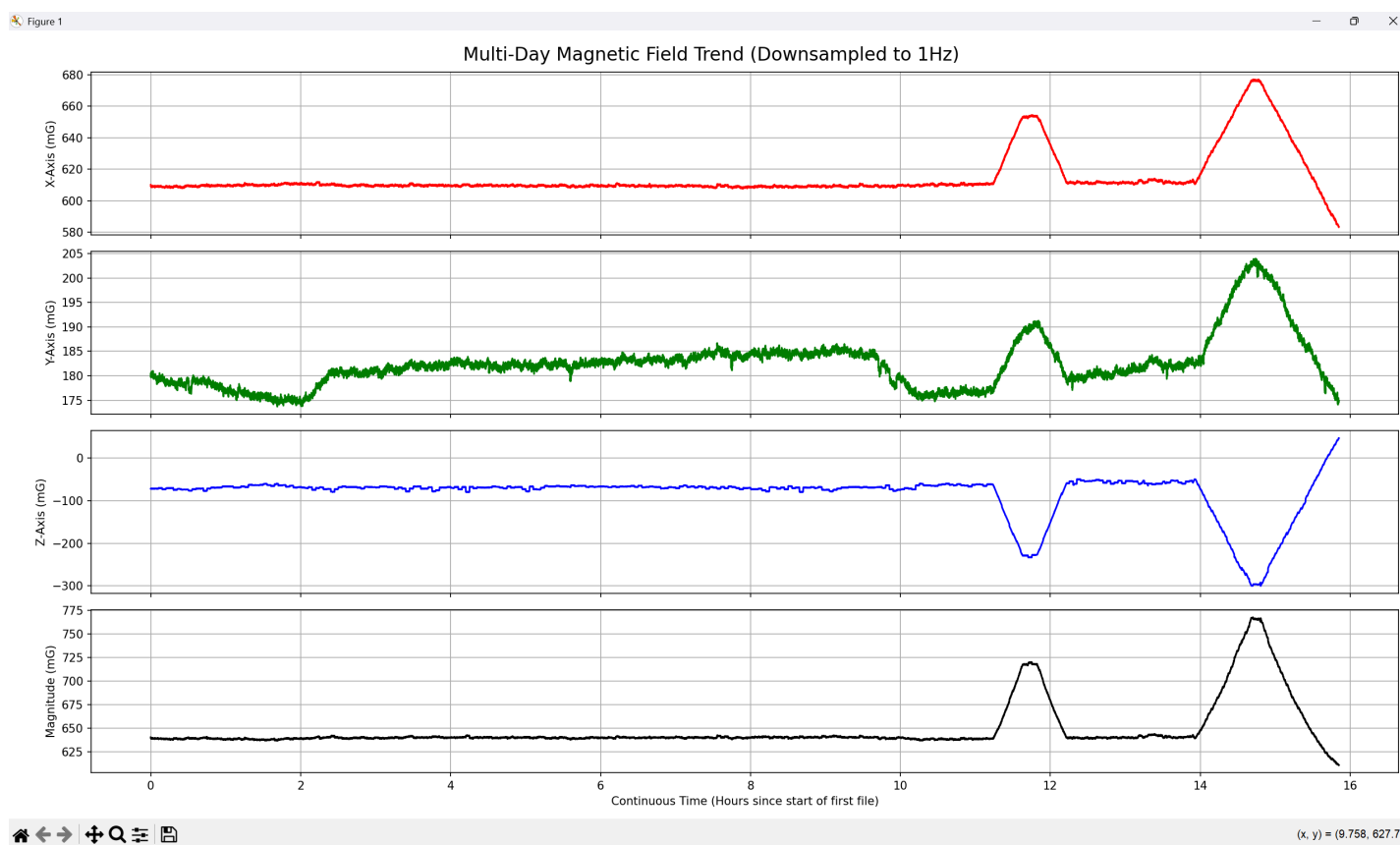
This is disturbance from other lab, none of lab members were near the table this time



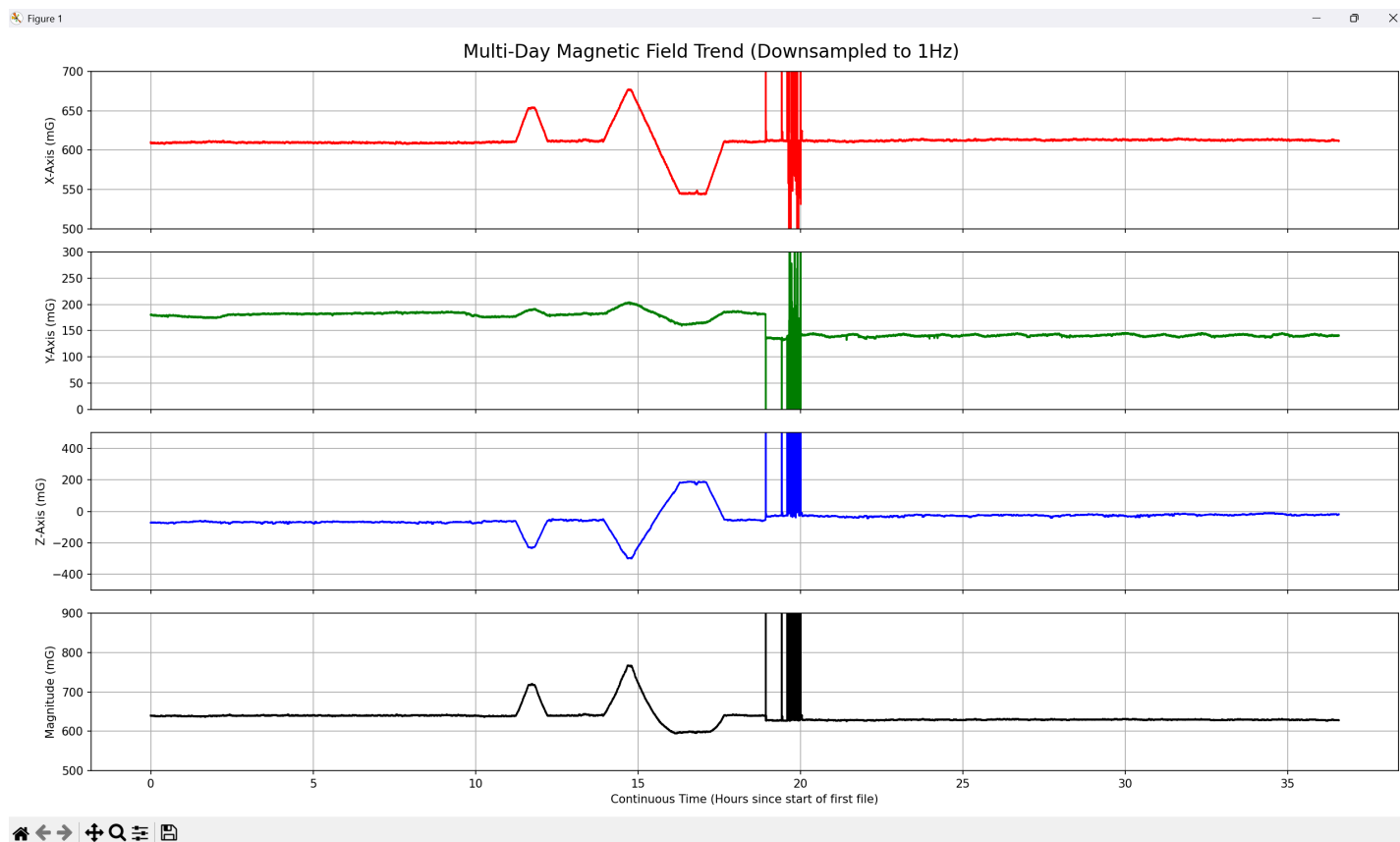
Saturday sunday



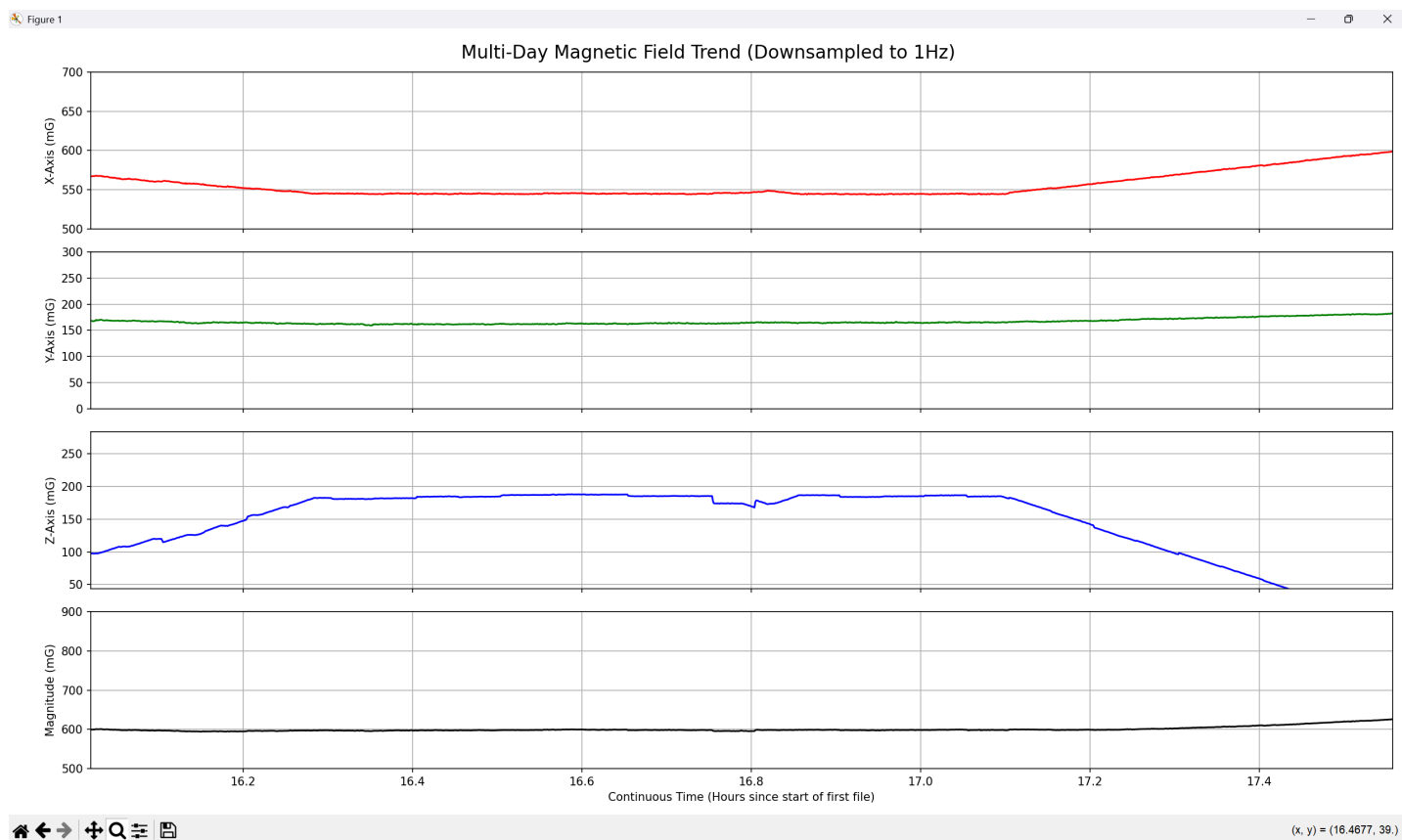
Monday tuesday



Tuesday, no coils have ever been turned on yet, clear b field from other lab. It crosses zero point in z-axis, might be an issue since TDK lambda is single side, might require H bridge



Tuesday and Wednesday. 218.3mG/H in X axis, 954mG/H in Z axis

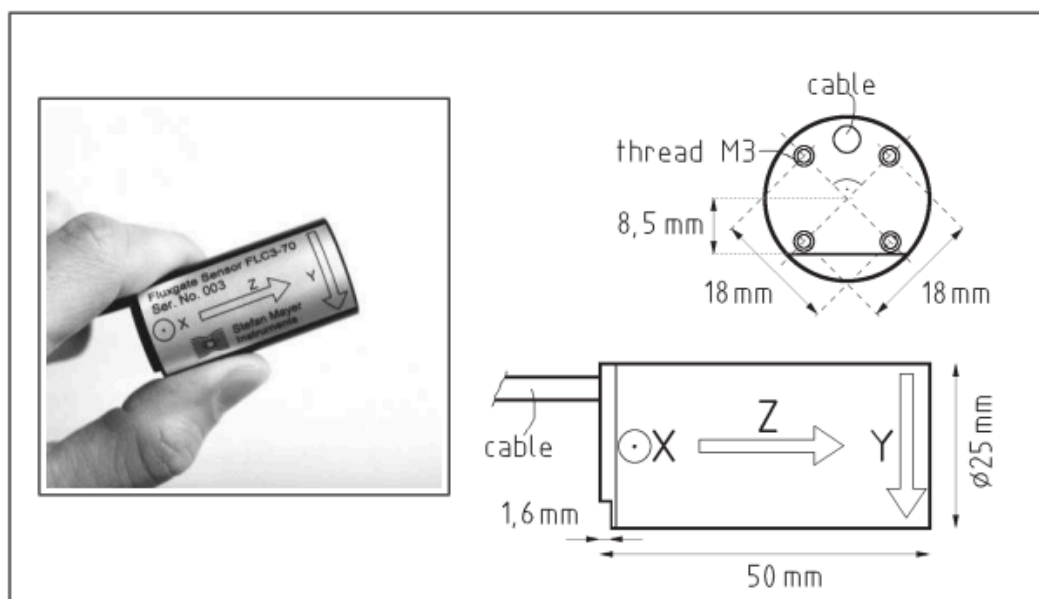
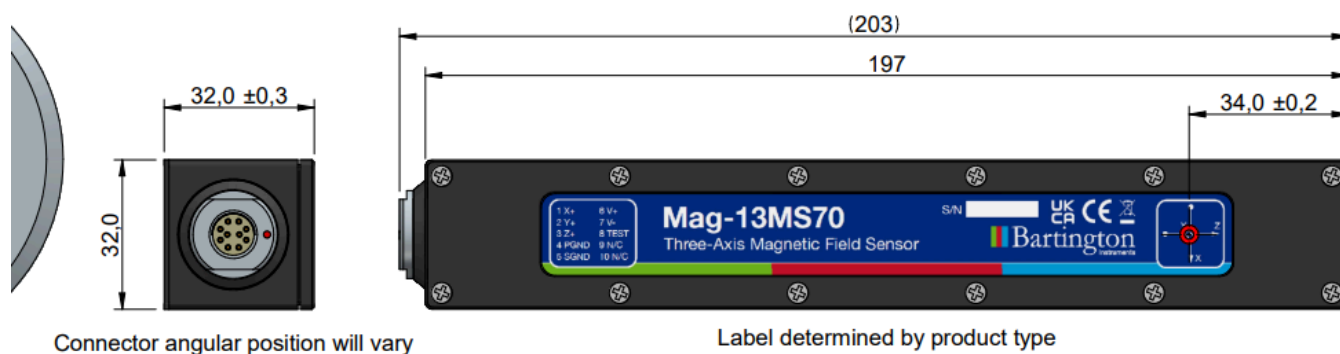


Current ideas:

1. use 4 sensors and take mathematical average, but orthogonality of axes issue
2. use 1 sensor and make handshake every once in a while with atom spectroscopy
 - a. proof of concept by using the 2 adafruit sensor and doing handshake process
 - b. dont need handshake everytime, just handshake calibration once
3. run atom spectroscopy multiple times, narrow down on the actual zero field

	Base Noise Density	Noise at 10 Hz	Noise at 100 Hz	Noise at 1 kHz
3 (Ultra-	5 pT/√Hz (0.05 μG/√Hz)	~0.16 μG RMS	0.5 μG RMS	~1.58 μG RMS
lini	~50 pT/√Hz (0.5 μG/√Hz)	< 5.0 μG RMS (per datasheet)	~5.0 to 7.0 μG RMS	~15.8 μG RMS
tive)	300 pT/√Hz (3.0 μG/√Hz)	~9.5 μG RMS	30.0 μG RMS	~95.0 μG RMS*

- Twinleaf: Volume: 16×16×87 mm³



- Barington price

scan each axis, check the paper

Our RF signal is generated by a function generator, sent to a circular coil attached to the bottom view port, which provides an oscillating magnetic field along the z axis. We calibrate three coils independently by scanning the voltage of one coil and recording the resonant RF frequency. The results are fitted by the function $B_i = \sqrt{(k_i V_i + b_i)^2 + B_{res}^2}$, where i denotes x, y, z , V is the input voltage for PID. We minimize the residual magnetic field when calibrating the x and y bias fields, while we retain a finite offset along y when calibrating the z bias field. Typical fittings for y and z coils are shown in Figure 2.16. We found that flip the direction of the field during calibration (to make fitting curves a "V" shape) could significantly reduce the fitting error and help us determine the zero crossing point of the magnetic field.

